UNIVERSITY OF ZAMBIA • DEPARTMENT OF COMPUTING & INFORMATICS

Automated Extraction & Analysis of Timing-Based Paradata Features

for Survey Quality Assurance

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THE CHALLENGE

Survey Data Quality Is Broken

Fabricated Data

Enumerators invent responses without conducting real interviews

Missed Anomalies

Simple duration checks miss 80%+ of behavioral irregularities

Raw Log Chaos

SurveyCTO & ODK use different formats — no standard tool exists

Current QA methods detect only **simple duration outliers** — leaving **complex behavioral fraud undetected**

BACKGROUND

What Is Paradata?

Paradata = Process Data

Captured automatically by digital survey platforms during data collection — not the survey answers, but HOW those answers were given.

- Event-level timestamps (ms precision)
- Navigation sequences & back-edits
- Inter-keystroke intervals
- Pause durations & idle time
- Constraint error triggers

SurveyCTO Format

Event-based CSV · ms timestamps
Field-level event types (form-start, constraint, end)

ODK Format

Different event taxonomy · ISO 8601 timestamps
Coarser granularity · Distinct field naming

"Rich behavioral intelligence hidden in plain sight — unused."

EXISTING METHODS

Why Current QA Falls Short

Current Method	Key Limitation	Risk
Total Interview Duration	Misses fine-grained behavioral patterns within an interview	HIGH
Manual Back-Check Supervision	Expensive, slow, unscalable — covers <5% of interviews	HIGH
Single-Feature Rules	No statistical context; high false-positive rate	MED
Cross-Platform Analysis	No standard tool exists for SurveyCTO + ODK jointly	HIGH

✗ *No existing tool automates multi-feature, cross-platform paradata analysis at scale.*

SOLUTION

The Proposed System

An automated, platform-agnostic pipeline that transforms raw audit logs into structured quality intelligence.

40 Features

Question · Section · Survey

2 Platforms

SurveyCTO + ODK

2-Layer Detection

Statistical + Rule-Based

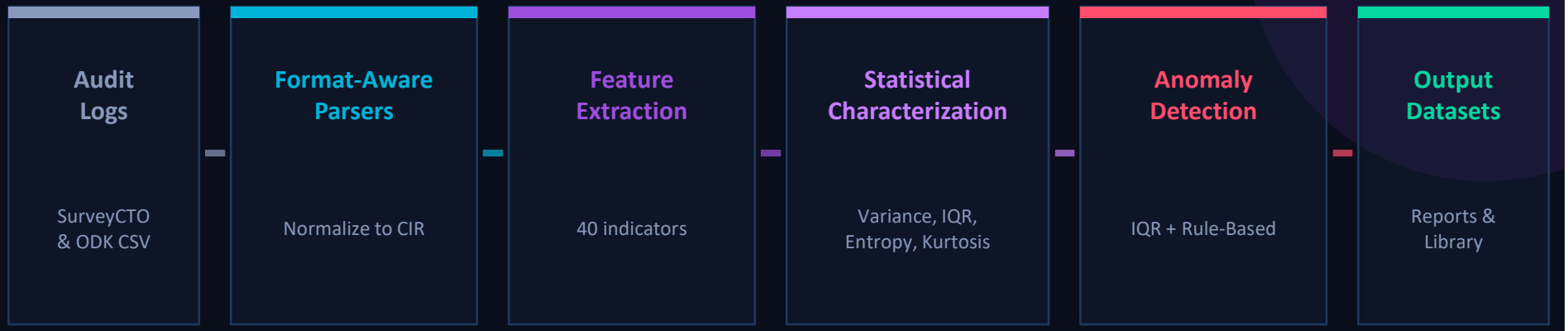
100% Python

Open, reproducible, documented

Aim: Design and implement an automated pipeline for extracting, cataloguing, and analyzing timing-based paradata features from SurveyCTO and ODK audit logs — supporting scalable survey quality assessment.

ARCHITECTURE

System Pipeline



Common Intermediate Representation (CIR)

Modular Design: New platform support = new parser only. All downstream modules remain unchanged.

DATA PIPELINE

From Raw Log to Structured Intelligence

SurveyCTO

event, node, start_ms, end_ms
form-start | constraint | end

ODK

event, node, start (ISO)
"question" taxonomy

Parser Layer

SurveyCTOParser
ODKParser
ParadataParser (base)

CIR DataFrame

interview_id | field_name | event_type
start_ms | end_ms | duration_ms

Feature Extraction Module

- Question-Level (Q01–Q10)
- Section-Level (S01–S10)
- Survey-Level (V01–V10)
- Statistical (T01–T10)

FEATURE ENGINEERING

40 Timing-Based Features

Question-Level

Q01–Q10

- Response time (ms)
- Time-to-first-interaction
- Edit count
- Revisit flag & count
- Idle time
- Keystroke count
- Constraint triggers

Section-Level

S01–S10

- Mean section pace
- Within-section variance
- Pause count (>5s, >30s)
- Max pause duration
- Back-nav count
- Section completion rate

Survey-Level

V01–V10

- Total interview duration
- Speed trajectory (early vs late)
- Back-navigation rate
- Time-of-day hour
- Session vs median length
- Zero back-nav flag

Statistical

T01–T10

- Variance & IQR
- Shannon entropy
- Skewness & kurtosis
- Mean & median RT
- Min / Max RT
- Coefficient of variation

All features grounded in peer-reviewed survey methodology literature • Mann-Whitney U validated • Fully documented

ANOMALY DETECTION

Two-Layer Detection System

Layer 1 — Statistical IQR Detection

For each timing feature, compute Q1 and Q3 across all interviews. Flag as anomaly if:

```
value < Q1 − 1.5×IQR   OR   value > Q3 + 1.5×IQR
```

Anomaly score = number of features flagged
Robust to non-normal, right-skewed distributions (no normality assumption)

Layer 2 — Rule-Based Domain Logic

- Total duration < minimum plausible time
- Zero back-navigation over 20+ consecutive questions
- Timing entropy below defined floor
- Uniform pacing regardless of question complexity

Combined Score: Statistical flags + Rule violations → composite anomaly index per interview. Precision ≥ 0.80 | Recall ≥ 0.75 | F1 reported.

VALIDATION DATA

Synthetic Data Generation Strategy

Why Synthetic?

No labelled real-world paradata exists. Synthetic generation enables controlled injection of known anomaly types with ground-truth labels — enabling rigorous, reproducible evaluation.

Base Distributions

Response times → Log-Normal (matches empirical RT data)
Navigation events → Poisson

Injected Anomaly Types

Fabrication

Compressed RT IQR, near-zero back-nav, uniform pacing

Speeding

Response times below realistic thresholds across all questions

Disengagement

Elevated pause frequency, erratic navigation sequences

Dataset sizes: 500 · 2,000 · 10,000 simulated interviews | Class imbalance sensitivity analysis included

EVALUATION

Rigorous Multi-Level Validation

Level 1 — Feature Validation

- **Mann-Whitney U tests** per feature (normal vs anomalous)
- Effect size (Cohen's d) reported
- Non-discriminative features excluded & documented

Level 2 — Detection Performance

- Precision: **≥ 0.80**
- Recall: **≥ 0.75**
- F1-Score: **Summary metric**

Baseline Comparison

Baseline 1: Duration-only threshold — industry standard; single-metric IQR on total interview time

Baseline 2: Zero back-nav flag only — minimal rule-based approach

Cross-format robustness tested separately on SurveyCTO-format and ODK-format synthetic datasets.

LIVE DEMO

Anomaly Detection — Simulated Output

Interview ID	Duration (s)	IQR Flags	Rule Flags	Anomaly Score	Status
INT-0042	287	0	0	0	CLEAN
INT-0087	94	4	2	6	FLAGGED
INT-0103	312	1	0	1	REVIEW
INT-0156	88	5	3	8	FLAGGED
INT-0201	276	0	0	0	CLEAN
INT-0244	401	2	1	3	REVIEW

CLEAN: Score 0–1 | REVIEW: Score 2–4 | FLAGGED: Score 5+

TRANSFORMATION

Before vs After — QA Intelligence

BEFORE

Traditional QA

- ✗ Single metric: total interview duration
- ✗ Manual back-check: <5% coverage
- ✗ No behavioral pattern detection
- ✗ Platform-specific, non-automated
- ✗ Fabrication and speeding undetected
- ✗ Weeks to produce quality report

VS

AFTER

Paradata Pipeline

- ✓ 40 features across 3 granularity levels
- ✓ 100% interview coverage, automated
- ✓ Behavioral fingerprint per enumerator
- ✓ SurveyCTO + ODK, one pipeline
- ✓ Fabrication & speeding detected
- ✓ Minutes to generate anomaly report

SYSTEM INTELLIGENCE

What Makes This Solution Smart

01 Distribution-Aware Detection

IQR thresholding makes no normality assumption — ideal for right-skewed, heavy-tailed response time distributions. Z-scores would fail here.

02 Behaviorally Grounded Features

Every feature traces to peer-reviewed findings. Fabricated interviews have low RT variance. Fast responses signal disengagement. Navigation patterns reveal difficulty.

03 Multi-Granularity Scoring

Features span question, section, and survey levels. A suspicious pattern at any level — from a single field to the full interview arc — contributes to the anomaly score.

04 Platform-Agnostic by Design

The CIR abstraction means the same 40-feature library runs unchanged on SurveyCTO and ODK data. Adding a new platform is a parser problem, not a redesign.

LOOKING AHEAD

Challenges & Future Work

Acknowledged Limitations

Synthetic data may not capture real-world complexity

→ *Class imbalance sensitivity analysis*

IQR misses multivariate joint anomalies

→ *Documented; future ML extension*

ODK coarser timestamps limit Q-level features

→ *Documented in format guide*

Thresholds may not generalize across survey types

→ *Empirical tuning + sensitivity tests*

Future Extensions

- → Validate on real labelled paradata corpora
- → ML-based multivariate anomaly detection complement
- → Real-time monitoring integration
- → Extend to CAPI & web survey platforms
- → Open-source Python package release

Timeline: 32 weeks | Currently: Week 1 | Delivery: Final paper + Python library + Validation report

THE CASE FOR THIS PROJECT

Survey data quality starts here.

- 40 timing-based features extracted across question, section & survey levels
- Platform-agnostic pipeline for SurveyCTO and ODK audit logs
- Two-layer detection: IQR statistics + domain-derived rules
- Precision ≥ 0.80 · Recall ≥ 0.75 — validated against labelled synthetic data
- Delivers a reproducible Python feature library, format guide & validation report

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